

ZEISS PENTERO®*

Microscope Interface for Stealth AXiS™ Surgical Systems with Cranial Quick Reference Guide

Description

The Medtronic Navigation microscope tracker is an accessory to the Stealth AXiS™ Surgical System. It consists of a tracker assembly that attaches to the ZEISS PENTERO microscope.

Intended use

The Medtronic Navigation microscope tracker is an accessory to the Stealth AXiS™ Surgical System. It is intended to be used with the ZEISS PENTERO microscopes and the Stealth AXiS™ Surgical System to enable microscope tracking positions during surgical procedures.

Indications for use

The Stealth AXiS™ Surgical System is intended for precise positioning of surgical instruments and as an aid for precisely locating anatomical structures in open, minimally invasive, and percutaneous procedures. The Stealth AXiS™ Surgical System is indicated for medical conditions in which the use of stereotactic surgery may be appropriate, and where reference to a rigid anatomical structure, such as the skull or vertebra, can be identified relative to a CT or MR based model, fluoroscopy images, or digitized landmarks of the anatomy.

Contraindications

None

Material composition

Material contained in product that can cause an allergic reaction: Nickel

For additional materials of concern information such as REACH, CA Prop 65, or other product stewardship programs, go to www.medtronic.com/productstewardship.

Compatibility

This device is compatible with Medtronic Stealth AXiS™ computer-assisted surgery systems and Medtronic navigation instruments.

System level manuals

This device can be used with Medtronic computer-assisted surgery systems. See the Stealth AXiS™ system level manuals and clinical application for descriptions of patient groups, intended users, potential clinical benefits, side effects, and potential complications.

Qualified microscopes

When used with the Stealth AXiS™ Surgical System with Stealth AXiS™ Cranial clinical application, the Medtronic Navigation microscope tracker is qualified with the following ZEISS PENTERO microscopes: ZEISS OPMI PENTERO 900

Warnings

Warning: Inspect the microscope tracker for damage and cleanliness before use. If visibly damaged or bent, do not use the microscope with the Stealth AXiS™ Surgical System. Do not attempt to repair it.

Warning: Use care when handling the microscope. Do not allow the microscope tracker to collide with anything.

Warning: If the microscope tracker assembly is damaged during the procedure, abort navigation and abandon use of the microscope with the Stealth AXiS™ Surgical System.

Warning: Frequently confirm localization accuracy and system responsiveness during navigation. If accuracy degrades, re-register the patient. Refer to the Stealth AXiS™ Cranial clinical application for instructions on how to re-register the patient.

Warning: The device is designed for re-use and is supplied non-sterile.

Warning: Use caution and exercise care in order to prevent injury during use. See the corresponding system level manuals and clinical application for further handling instructions.

Warning: To avoid potential exposure to blood-borne pathogens and chemicals, use appropriate Personal Protective Equipment when handling, processing, or disposing of Medtronic devices.

Warning: Confirm navigational accuracy via anatomical landmarks and accuracy checkpoints before navigation.

Warning: When navigating with the microscope, keep the crosshair in the same ocular that was used to calibrate the microscope. Changing which ocular has the crosshair can cause inaccurate navigation and result in patient injury or death.

Precautions

Caution: Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.

Caution: Refer to the Stealth AXiS™ system level manuals and clinical application for additional instructions, warnings, and cautions.

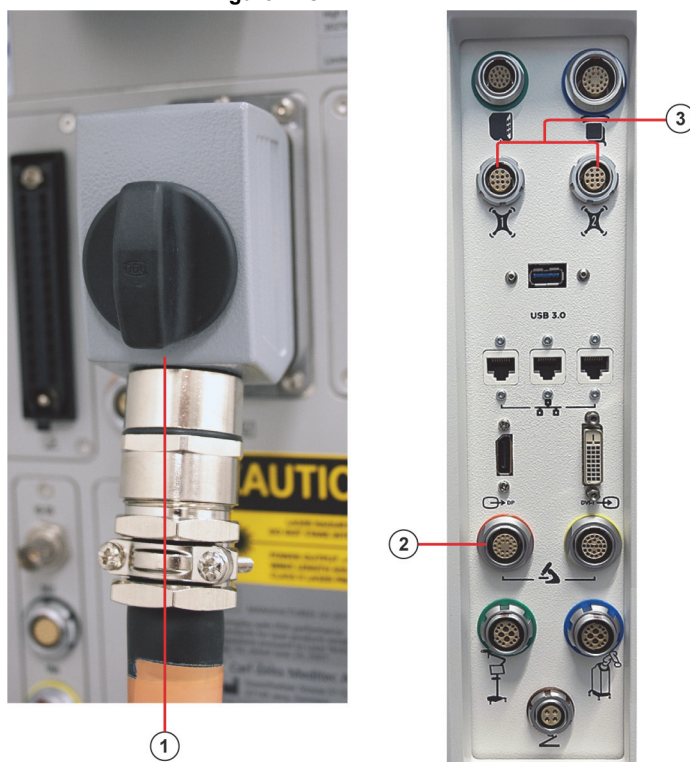
Setting up equipment

1. Connect the Stealth AXiS™ Surgical System to a power source and turn it on.
2. Connect the ZEISS OPMI PENTERO 900 microscope to a power source and turn it on.

Note: Do not connect to an external display device while performing a clinical procedure. When connecting or disconnecting an external monitor, all monitors on the system will go black for a few seconds while the video subsystem reconfigures.

- a. Connect the large, rectangular ODU™* connector of the Stealth AXiS™ Surgical System microscope communication cable to the back of the ZEISS OPMI PENTERO 900 **(1)** microscope.
- b. Connect the orange latching push-pull LEMO™* connector on the other end of the Stealth AXiS™ Surgical System communication cable to the Microscope port **(2)** on the instrument interface panel of the Stealth AXiS™ Surgical System.
- c. Connect the gray latching push-pull LEMO™* connector on the other end of the Stealth AXiS™ Surgical System communication cable to Instrument port 1 or 2 on the instrument interface panel of the Stealth AXiS™ Surgical System **(3)**.

Figure 1: Connect the cables



- 1 ZEISS OPMI PENTERO 900 ODU™ connector
 - 2 Microscope port (analog) on Stealth AXIS™ Surgical System
 - 3 Instrument ports 1 and 2 on Stealth AXIS™ Surgical System
3. Drape the microscope using an appropriate ZEISS OPMI PENTERO 900 microscope drape. Refer to the instructions for use that accompany the ZEISS PENTERO microscope for proper draping.
 - a. Pull the drape taut over the LED tracker on the microscope head.
 - b. Secure the drape in place with sterile rubber bands.

Note: Make sure that the LEDs are clearly visible and unobstructed by the rubber bands and any wrinkles in the drape.
 4. Set up tracking.
 - a. In the **Equipment** task in the clinical application, make sure that the microscope is connected to the Stealth AXIS™ Surgical System.

If the microscope equipment card is not displayed, search for the microscope in the **Add Equipment** list, and tap **[+]** next to the microscope equipment card.

If the microscope equipment card indicates that the microscope is not connected to the clinical application, refer to the “Troubleshooting” section.

Note: If you have two or more active instruments in a procedure, assign each instrument to an instrument port. See the clinical application instructions for use for information about how to assign an active instrument.

- b. Position the camera so that both the microscope tracker and patient reference frame display in the tracking view on the clinical application. Make sure that the microscope tracker is the only tracked instrument in the camera’s field of view to avoid navigational interference.

Navigating

Warning: Before localizing, verify navigational accuracy of the system. If navigational accuracy is affected, abort use of the system.

Note: Video injection is not supported.

1. Verify navigational accuracy of the system by comparing points of interest on the clinical application to the corresponding points of the patient anatomy.
2. Localize the microscope.

To “localize” means to identify the location of the tip of a Stealth AXiS™ Surgical System probe on the preoperative patient images in the clinical application. When you use the microscope to localize, this “tip” is the point you observe at the intersection of the ocular crosshairs in the focal plane of the microscope.

- a. Focus the microscope on the anatomy of interest.
- b. Observe the updated localization site in the images on the clinical application.

Note: As you decrease magnification, the depths of field (DOF) increases. A larger DOF will result in objects further above and below the navigated focal point appearing in focus. To narrow the DOF around the navigated focal point, increase microscope magnification.

3. Focus on a reference frame divot.

Center the ocular crosshair or converging lasers (if equipped) on a reference frame divot, increase magnification to the highest level, and focus the microscope on the bottom of the divot.

- Alternatively, you may focus on the center of the ocular crosshair on the etched divot of the reference frame.
- Confirm that the banner message indicates that the distance to the divot is less than 2 mm.

- The localization site is at the intersection of the crosshair installed within the eyepiece (crosshair reticle). Make sure that the crosshair reticle is installed in the microscope. If not, remove the regular eyepiece and replace it with the crosshair reticle eyepiece.

Warning: If inaccuracy is detected, the tracker assembly may have been moved or repositioned. Contact Medtronic Navigation to schedule a microscope install calibration. Calibration of the microscope tracker must be performed by a trained service person.

Troubleshooting

Locate the issue in the following table and follow the action steps. If the symptom persists, contact Medtronic technical services for assistance.

Issue	Action
The Equipment task shows no communication between the Stealth AXiS™ Surgical System and the microscope. Or The Equipment or Instruments task shows the microscope tracker as disconnected.	1. Make sure that the microscope is powered on. 2. Check the cabling path between the Stealth AXiS™ Surgical System and the microscope: a. Make sure the ODU™ connector is attached to the microscope. b. Make sure that the orange LEMO™ connector is connected to the Microscope port on the instrument interface panel of the Stealth AXiS™ Surgical System. c. Make sure that the gray LEMO™ connector is connected to Instrument port 1 or 2 on the instrument interface panel of the Stealth AXiS™ Surgical System. d. If you are using multiple active instruments in a procedure, open the Instruments task and verify that the clinical application registers that the microscope is plugged in to the correct Instrument port. If the clinical application registers that the microscope is plugged into the wrong Instrument port, re-assign the microscope.

Issue	Action
The microscope displays a red status bar in the Navigation task even though all cables are properly connected.	Position the camera so that both the microscope tracker and patient reference frame are in the field of view. For optimal positioning, align the camera to the forward right of the surgical area with the microscope tracker in the camera's field of view. Confirm that the drape is positioned properly over the tracker. If the problem persists, contact Medtronic technical services with the part number and serial number of the microscope tracker. The probe geometry error may be too high to use the microscope with the navigation clinical application.

Disconnecting the interface

Leave the microscope LED tracker permanently attached to the head of the microscope. This helps prevent invalidating the calibration.

Warning: Do not modify, disassemble, or remove the microscope tracker. Removing and reattaching the microscope tracker changes the position of the tracker in relation to the microscope. This invalidates the calibration of the tracker, and a new calibration is required to prevent inaccurate navigation. Calibration of the microscope tracker must be performed by a trained service person.

1. Disconnect the microscope communications cables from the Stealth AXiS™ Surgical System instrument interface panel.
2. Disconnect the microscope communication cables from the microscope.
3. Coil and secure the cables on the microscope using Velcro™ ties.

Maintenance

1. Make sure that the microscope tracker is securely mounted to the microscope. If the microscope tracker is not securely mounted, contact Medtronic Navigation.
2. Clean the microscope tracker.
 - a. Refer to Table 1 for compatible disinfectants and recommended chemistries.

Table 1: Compatible cleaner and disinfectant chemistries

Disinfectant type	Disinfectant level
Quaternary ammonium germicidal detergent solution	Low level
Phenolic germicidal detergent solution	Low level
Isopropyl alcohol	70%
Sodium hypochlorite	Low level (5.25% to 6.15%)

- b. Wipe and clean the external surfaces of the microscope tracker with a moistened low-level disinfectant wipe per the disinfectant manufacturer's instructions. If a premoistened wipe is not available, use a clean wipe and moisten with the disinfectant. Do not over saturate the wipe so much that liquid can enter any openings.

Warning: Prevent fluid from entering any openings on the microscope tracker and cable.

- c. Wipe the external surfaces with a clean wipe moistened with water or 70% isopropyl alcohol to remove disinfectant residue on the microscope tracker.
 - d. Allow the microscope tracker to air dry completely before reusing.
 - e. Visually inspect the microscope tracker for cleanliness. If visible soil or residue remains, repeat the steps to clean the microscope tracker.
3. Visually inspect the microscope tracker for damage.

Make sure that the microscope tracker does not have any of the following indications of damage:

- Illegible etchings or markings
- Visible cracking or bending
- Discoloration, corrosion, stains or rust

If visibly damaged, the microscope tracker is at the end of its useful life and should not be used with the Stealth AXIS™ Surgical System.

4. Clean the microscope interface cables.
 - a. Refer to the following table for compatible disinfectants and recommended chemistries for the microscope interface cables.

Table 2: Microscope interface cable disinfectants

Disinfectant type	Disinfectant level
Quaternary ammonium germicidal detergent solution	Low level
Phenolic germicidal detergent solution	Low level
Isopropyl alcohol	70%
Sodium hypochlorite	Low level 5.25% to 6.15%

- b. Power down and unplug the system before performing cleaning or disinfection procedures.
- c. Unplug external cables.
- d. Clean the cable by wrapping a low-level disinfectant wipe completely around the cable starting at one connector.

Warning: Do not allow liquid to enter the interior surfaces of the connector.

- e. Wipe the cable from end to end by pulling the cable through the wipe. Ensure contact with the connector, bend relief, and cable.

- f. If disinfection is desired, make sure that the surface to be disinfected remains in contact with the disinfectant for the specified contact time as directed by the manufacturer.
 - The contact time refers to the minimum amount of time the surface needs to remain visibly wet.
 - Use additional wipes, if needed, to maintain continuous contact time with the disinfectant.
- g. Examine the surfaces for visible soil. If soil is present, repeat cleaning.
- h. Wipe the external surfaces with a clean wipe moistened with water or 70% isopropyl alcohol to remove disinfectant residue.

Specifications

Table 3: Microscope tracker specifications

Operating temperature	18°C to 30°C (64°F to 86°F)
Operating Relative humidity	30% to 75% Non-condensing
Storage temperature	10°C to 40°C (50°F to 104°F)
Storage Relative humidity	10% to 80% Non-condensing

Storage

Ensure equipment and packaging are dry before storing. Store in dry, clean conditions at ambient room temperature such that the equipment and packaging are not compromised.

Disposal

When the device shows visible signs of wear, it is at the end of its useful life. Dispose of it in accordance with your facility's procedure for medical devices and national regulations.

Contact information

If a serious incident occurs in relation to the use of this device, report it to Medtronic Navigation. If a serious incident occurs in the European Union, also report it to the competent authority in the Member State where the incident occurred.

Assistance

For questions about the device or to order accessories, contact your Medtronic Navigation representative or call Medtronic Navigation, Inc. at +1 800 595 9709 in the USA or +1 720 890 3160 for technical support.

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This product may be protected by US Patents listed at: medtronic.com/patents.

Symbol definitions

The following symbols may appear on device packaging:

Rx Only

Caution: Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.



Do not dispose of this product in the unsorted municipal waste stream. Dispose of this product according to local regulations. See <http://recycling.medtronic.com> for instructions on proper disposal of this product.



China RoHS compliant: Environmental protection use period of 50 years.



R_x Only

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